

Clinical Decision Making & Pattern Recognition in Healthcare

Topic Definition

At its core, Clinical Decision Making and Pattern Recognition in Healthcare is about giving machines the ability to do what experienced clinicians and auditors do naturally — spot something that doesn't look right. The difference is scale. A trained reviewer might catch a handful of suspicious claims in a day; a well-built **machine learning system** can screen thousands in seconds. The **TPO framework** — Treatment, Payment, and Operations — provides a useful way to think about where that capability creates real value (Shafqat et al., 2020). It covers everything from predicting which patients need intervention, to detecting fraudulent billing, to flagging when a hospital's admission volume spikes in ways that don't add up.

For this project, I focused on the **Payment** lane, specifically the problem of detecting outlier insurance claims. The technique I used is **Isolation Forest** — an unsupervised algorithm that works by asking a simple question: how easy is it to isolate this data point from everything else? Normal claims sit in dense clusters and take many steps to separate out. Anomalous ones are easier to isolate because they sit in sparse, unusual space (Liu et al., 2008). That intuition turns out to be surprisingly powerful when applied to healthcare billing data.

Relevant Trends

- ▶ The financial stakes are hard to ignore. Healthcare fraud, waste, and abuse costs the U.S. system an estimated \$300 billion every year (FBI, 2024), and the DOJ recovered over \$2.9 billion in False Claims Act settlements in FY2023 alone (DOJ, 2024). Payers aren't just under pressure to recover money after the fact anymore — they're expected to show they have proactive systems in place.
- ▶ Explainable AI is quietly solving one of the biggest blockers to ML adoption in healthcare. Frameworks like SHAP make it possible to not only flag an anomalous claim, but explain exactly why it was flagged — which features drove the score and by how much (Lundberg & Lee, 2017). For compliance teams, that distinction between a black-box alert and an auditable finding is everything.
- ▶ Agentic AI is starting to change what's possible end-to-end. Instead of just flagging a claim and stopping, systems can now chain detection, explanation, and report generation into a single automated workflow — shifting the whole operation from post-payment recovery toward real-time review before the money ever leaves.

Opportunities & Threats

Opportunities

- ▶ The financial stakes matter. Healthcare fraud costs the U.S. an estimated \$300 billion annually (FBI, 2024), and the DOJ recovered \$2.9 billion in False Claims Act settlements in FY2023 alone (DOJ, 2024). Payers aren't just expected to recover money anymore — they need to show they have proactive systems preventing it.
- ▶ Anomaly rates aren't uniform across insurers — and that's a finding on its own. My analysis showed some payer-condition combinations carry significantly higher risk than others. Blanket review thresholds miss this. Targeted stratification is far smarter.
- ▶ SHAP turns a black-box alert into something a reviewer can act on. Each flagged claim comes with a feature-level breakdown of what looked unusual — which makes the workflow more efficient and creates a defensible compliance trail (Lundberg & Lee, 2017).
- ▶ The architecture extends naturally across TPO. Length-of-stay drift maps to Treatment; admission volume spikes map to Operations. One well-built detection system can serve all three lanes from a single codebase.

Threats

- ▶ Fraud patterns evolve. Bad actors watch for detection systems and adapt — which means a static model needs regular retraining to stay relevant. That's a real operational cost to plan for.
- ▶ False positives are painful. Wrongly flagging a legitimate claim creates provider friction that's hard to repair. Getting the precision-recall balance right has real relationship consequences, not just technical ones.

- ▶ Regulation is still catching up. CMS requirements around algorithmic transparency and appeal rights for automated claim decisions are evolving — and compliance obligations vary significantly by state.

Strategic Recommendations:

- 1. Build a prepayment scoring layer.** The highest-leverage change Cotiviti could make is moving anomaly detection upstream — scoring claims before payment releases rather than investigating after the fact (Liu et al., 2008). My proof-of-concept shows this is technically straightforward: a 200-tree Isolation Forest on contextual billing features processes 10,000 records in seconds (Prasad, 2023). The hard part is integration, not the model — but catching overpayments before they happen rather than clawing them back through dispute processes justifies that investment.
- 2. Pair every flag with a SHAP explanation.** A claim flagged with no explanation creates more work, not less. SHAP values solve this by showing exactly which features drove the score (Lundberg & Lee, 2017) — so every flag arrives with a note like: *"Billing of \$24,300 for arthritis under Aetna is 4.1 standard deviations above expected, and the one-day stay doesn't match the billed complexity."* That's something a reviewer can act on immediately, and it creates the documented audit trail that compliance teams and regulators increasingly require.
- 3. Turn the heatmap into a client intelligence product.** What I found most interesting was that insurer-level anomaly rates tell their own story — some payer relationships are consistently higher risk for the same conditions. If Cotiviti has surfaced that as a live dashboard, it would function as an early-warning system for emerging fraud patterns (OIG, 2023). That's the shift from audit service to intelligence platform — a meaningfully different value proposition.

Conclusion

Building this POC clarified something easy to miss: the hardest part of healthcare anomaly detection isn't the algorithm. Isolation Forest is well-understood, SHAP is accessible, and Python makes both straightforward to implement (Liu et al., 2008; Lundberg & Lee, 2017; Pedregosa et al., 2011). The real work was framing the problem correctly — recognizing that a \$15,000 claim is only suspicious in context, which is why engineering features like billing z-score and insurer-condition ratios mattered more than model tuning. The anomaly signals are in the data; they're just invisible without the right lens. Once visible, the path for any organization is clear: move detection upstream, make every flag explainable, and surface insurer-level trends as ongoing intelligence rather than a one-time audit output (Shafqat et al., 2020). That's how payment integrity becomes proactive.

References

- Department of Justice. (2024). False Claims Act settlements and judgments exceed \$2.9 billion in fiscal year 2023. U.S. Department of Justice. <https://www.justice.gov/opa/pr/false-claims-act-settlements-and-judgments-exceed-29-billion-fiscal-year-2023>
- Federal Bureau of Investigation. (2024). Health care fraud. U.S. Department of Justice. <https://www.fbi.gov/investigate/white-collar-crime/health-care-fraud>
- Liu, F. T., Ting, K. M., & Zhou, Z.-H. (2008). Isolation Forest. Proceedings of the 2008 IEEE International Conference on Data Mining (pp. 413–422). <https://doi.org/10.1109/ICDM.2008.17>
- Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems, 30. <https://arxiv.org/abs/1705.07874>
- Prasad, P. (2023). Healthcare dataset [Data set]. Kaggle. <https://www.kaggle.com/datasets/prasad22/healthcare-dataset>
- Shafqat, W., Byun, Y.-C., & Shafqat, S. (2020). Unsupervised machine learning approaches for anomaly detection in healthcare claims data. Healthcare, 8(4), 404. <https://doi.org/10.3390/healthcare8040404>